

Premature Death in California's Counties

By: Cameron Fong

1. Introduction

Premature death, measured as Years of Potential Life Lost (YPLL) per 100,000 population, is a fundamental indicator of population health that captures mortality occurring before the age of 75. Understanding the geographic distribution of premature death across counties is critical to identify health disparities, allocating public health resources, and designing spatially targeted interventions.

California with its 58 counties spanning diverse geographic, socioeconomic, and demographic contexts provides a compelling study area for spatial health analysis. Counties range greatly from densely urbanized coastal regions such as Los Angeles and San Francisco to the sparsely populated rural and agricultural inland counties such as Modoc and Trinity, creating substantial variation in both health outcomes and underlying risk factors.

Spatial statistical methods are well-suited to county-level data because health outcomes are rarely independent across geographic areas. Neighboring counties may share environmental exposures, healthcare infrastructure, socioeconomic conditions, and population behaviors, inducing spatial autocorrelation that violates the independence assumptions of classical regression. Failing to account for this spatial dependence can lead to biased coefficient estimates and subsequent inference.

The goal of this paper is to address the following question: what socioeconomic, behavioral, demographic, and environmental factors explain the spatial distribution of premature death rates across California counties, and do neighboring counties exhibit similar health outcomes? We hypothesize that higher poverty, unemployment, physical inactivity, and air pollution will contribute the most to increased premature death rates, and that these and other outcomes will exhibit positive spatial autocorrelation in which counties with high premature death rates cluster geographically near similarly affected neighboring counties.

To address this question, we fit and compare three models: an ordinary least square (OLS) regression as a non-spatial baseline, a proper conditional autoregressive (CAR) model estimated with maximum likelihood, and an improper CAR model estimated with Bayesian Markov Chain Monte Carlo (MCMC). The comparison of these models will allow us to assess whether accounting for spatial structure improves model fit and changes interpretation of predictor effects. We will also look at the choropleth maps of the outcome variable premature death rates and our chosen covariates to see how they reflect the coefficient results in our 3 models.

2. Data Description

Data for this analysis was obtained from the 2025 County Health Rankings (CHR) Analytic dataset produced annually by the University of Wisconsin Population Health Institute in partnership with the Robert Wood Johnson Foundation. The CHR dataset compiles county-level health outcome and factor measures for all counties in the United States from a variety of national data sources including the CDC, ACS, and BRFSS.

The analytic dataset was filtered to California's 58 counties, with state-level summary rows excluded. Variables were selected to represent 4 theoretical domains that could be key determinants of premature mortality: socioeconomic conditions, health behaviors, demographic composition, and environmental factors all of which were retained in their raw value form.

The outcome variable is the premature death rate as stated in the introduction weights early deaths more heavily than those that occur closer to 75, making it sensitive to most preventable causes of mortality. Predictors include: children in poverty rate (socioeconomic), unemployment rate (socioeconomic), physical inactivity rate (behavioral), percentage of population that is non-Hispanic (demographic), percentage of population that is Hispanic (demographic), percentage of population aged 65 and older (demographic), and food insecurity rate (socioeconomic). Additional candidate variables considered during screening included median household income, adult smoking rate, adult obesity rate, PM2.5 air pollution, percentage rural, poor physical health days, feelings of loneliness, insufficient sleep, and access to parks. County-level spatial boundaries were obtained from the United States Census Bureau via the *tigris* R package (2023 vintage).

3. Methods

3.1 Spatial Weights Construction

Spatial relationships among California's 58 counties were represented using a queen contiguity weights matrix, in which counties sharing a common border or corner vertex are defined as neighbors. The weights matrix W was row-standardized for use in the proper CAR model and retained in binary form for the Bayesian CAR model as required by the *CARBayes* package. The spatial weights object was constructed using the *poly2nb* and *nb2listw* functions from the *spdep* R package.

3.2 Moran's I

Global spatial autocorrelation in the premature death rate was assessed in the 3 models and the outcome variable using Moran's I statistic, which ranges from -1 (perfect dispersion) to +1 (perfect clustering), with values near 0 indicating spatial randomness. A two-sided test was conducted with a p-value threshold of 0.05.

3.3 Choropleth Maps

Choropleth maps were produced using quantile classification to visualize the geographic distribution of the outcome and all predictor variables across California counties. The maps were also produced to compare covariate results with the 3 models.

3.4 Variable Selection

An initial set of 17 predictor variables was assembled based on theoretical relevance to premature mortality across four domains: socioeconomic (children in poverty, unemployment rate, median household income, food insecurity, feelings of loneliness), behavioral (smoking, obesity, physical inactivity, insufficient sleep, poor physical health days), demographic (% non-Hispanic, % Hispanic, % aged 65+), and environmental/geographic (PM2.5 air pollution, % rural, access to parks). Counties with missing values on any retained variable were excluded from analysis, yielding a complete case sample of 58 California counties. A correlation matrix was first examined to identify pairwise relationships among predictors. A full OLS regression model was then fitted on the complete candidate set, and stepwise selection based on AIC using

bidirectional elimination was applied to identify an optimal subset of predictors. Multicollinearity among the retained predictors was assessed using Variance Inflation Factors (VIF), with values exceeding 10 considered indicative of serious multicollinearity and removed if there were any. Model adequacy was further evaluated through residual vs. fitted plots and normal Q-Q plots of residuals to confirm that OLS assumptions were reasonably satisfied prior to spatial modelling.

3.5 OLS Regression

An OLS regression model was fitted as a non-spatial baseline using the final variable set. Moran's I was applied to OLS residuals to formally test for residual spatial autocorrelation. A statistically significant result ($p < 0.05$) would indicate that the independence assumption is violated and that a spatial model is warranted. Model fit was assessed using AIC and BIC.

3.6 Improper CAR Model

The improper (intrinsic) CAR model was fitted using the S.CARleroux function from the CARBayes R package. The model takes the form:

$$Y_i = X_i\beta + \phi_i + \varepsilon_i, \quad \phi \sim \text{ICAR}(\tau^2), \quad \varepsilon_i \sim N(0, \sigma^2)$$

where ϕ is a vector of spatially structured random effects following the ICAR prior with precision parameter τ^2 . The spatial random effects ϕ capture residual spatial clustering in premature death rates not explained by the covariates. Posterior inference was conducted via Markov Chain Monte Carlo (MCMC) sampling with 200,000 burn-in iterations, 1,000,000 total samples, and a thinning interval of 10, yielding approximately 80,000 effective posterior samples. MCMC convergence was assessed using Geweke's diagnostic ($|z| < 2$ indicates convergence) and by inspection of trace plots.

3.7 Proper CAR Model

The proper CAR model was fitted using the spautolm function from the spatialreg R package. The proper CAR model introduces a spatial dependence parameter λ estimated from the data, where $\lambda = 0$ indicates no spatial structure (equivalent to OLS) and larger values indicate stronger spatial dependence among neighboring counties. The model takes the form:

$$Y = X\beta + \varepsilon, \quad \varepsilon \sim \text{CAR}(\lambda, \sigma^2)$$

where Y is the vector of premature death rates, X is the matrix of predictors, β is the vector of regression coefficients, and the error structure follows a proper CAR distribution parameterized by λ and variance σ^2 .

4. Results

4.1 Spatial Autocorrelation of Outcome

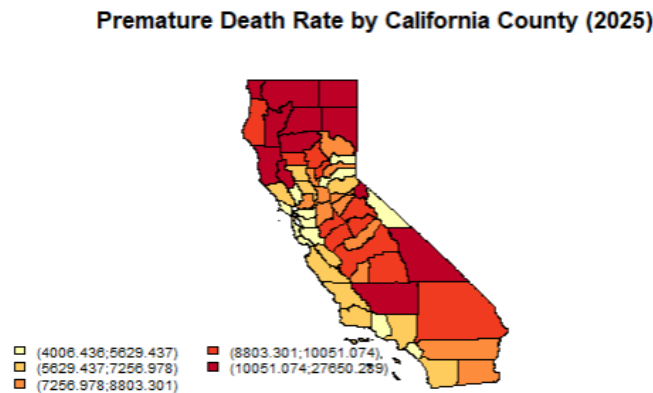


Figure 1

Figure 1 presents the choropleth map of premature death rates across California's 58 counties using quantile classification where the northern and inland sections of California have higher concentrated premature deaths as compared to coastal counties in the western and southern region exhibit lower rates. This suggests clustering of premature death rates and the potential for neighboring counties to have an impact with one another and motivates for formal testing for spatial autocorrelation.

The Moran's I test (Appendix figure 1) supports this theory with a p-value of 0.0201 representing that there is strong evidence of positive spatial autocorrelation in the outcome variable premature deaths.

4.2 Variable Selection

Following the variable selection techniques presented in section 3.4, a correlation matrix was created (Appendix Figure 2) prior to model fitting to identify pairwise relationships among predictors and the outcome to screen for potential multicollinearity. Children in poverty exhibit the strongest bivariate correlation with premature death with $r = 0.71$ and warrants interest in their addition to the final model. Other predictors are observed to have high collinearity with one another namely median household income and children with poverty with $r = -0.79$ which captures overlapping socioeconomic variation.

Stepwise AIC selection was then implemented on all 17 predictors of premature death or the full OLS model (Appendix figure 3), which then identified 7 predictors for inclusion in the final model (Appendix Figure 4): children in poverty, food insecurity, hispanic, non-hispanic black, physical inactivity, 65 or older, and unemployment rate. Our conclusions in the correlation matrix match these findings with variables of high collinearity such as median household income being excluded and those of high correlation with the outcome variable being kept. An ANOVA comparison (Appendix Figure 5) between the full and stepwise models confirmed that the dropped variables were not jointly significant with $p = 0.78$. VIF values for all retained predictors (Appendix Figure 6) ranged from 2.3 to 9 with no predictors exceeding the threshold of 10, indicating no serious multicollinearity concerns. Model adequacy is further confirmed by the residuals vs fitted plots (Appendix Figure 7) showing no clear pattern with points scattered

randomly around zero and the QQ-plot (Appendix Figure 7) shows residuals normally distributed on the theoretical line, indicating the model is appropriate.

Variable	N	Mean	SD	Min	Q25	Median	Q75	Max
Premature Death Rate (YPLL/100k)	58	8303.530	3553.813	4006.436	5795.242	8056.890	9709.310	27650.289
Children in Poverty (%)	58	0.162	0.057	0.061	0.116	0.157	0.206	0.295
Food Insecurity (%)	58	0.129	0.026	0.085	0.108	0.130	0.150	0.210
Hispanic (%)	58	0.329	0.185	0.064	0.164	0.280	0.478	0.863
Non-Hispanic Black (%)	58	0.030	0.028	0.004	0.011	0.019	0.041	0.130
Physical Inactivity (%)	58	0.223	0.042	0.128	0.192	0.218	0.250	0.323
65 and Older (%)	58	0.200	0.061	0.112	0.155	0.184	0.245	0.330
Unemployment Rate (%)	58	0.056	0.024	0.030	0.041	0.051	0.064	0.173

Table 1: Descriptive Statistics of Variables in Final Model

With the given table, we see that Table 1 presents descriptive statistics for all variables in the final model across California's 58 counties. The premature death rate exhibited the greatest absolute variation of any variable, ranging from 4,006 to 27,650 YPLL per 100,000 (mean = 8,304, SD = 3,554), with the most affected county experiencing nearly seven times higher premature mortality than the least affected. Among predictors, Hispanic population share showed the greatest relative variation (mean = 32.9%, SD = 18.5%, range: 6.4% to 86.3%), reflecting California's diverse demographic composition across counties. Children in poverty averaged 16.2% but reached 29.5% in the most disadvantaged counties, while unemployment exhibited a right skew (mean = 5.6%, median = 5.1%), suggesting a subset of counties with particularly high joblessness. Non-Hispanic Black population share showed limited county-level variation (mean = 3.0%, SD = 2.8%), which may constrain its explanatory contribution in regression models.

4.3 Choropleth Maps of Predictors

To contextualize the outcome map, choropleth maps of all predictor variables selected in variable selection were examined. Children with poverty (Appendix figure 8) appear more inland and towards the north where premature death rates were the highest, suggesting a potential socioeconomic gradient. Unemployment rate (Appendix Figure 9) showed a broadly similar geographic pattern, with elevated rates concentrated in northern and central inland counties coinciding with the regions of highest premature death, further reinforcing the role of economic deprivation in driving geographic disparities in premature mortality. Physical Inactivity (Appendix Figure 10) presented a more regionally heterogeneous pattern where the highest rates were concentrated in central and southern California and broadly overlapped with elevated premature death in those areas, though northern counties showed a weaker and less consistent correspondence, suggesting the relationship between physical inactivity and premature death may vary across regions of the state. Non-Hispanic Black (Appendix Figure 11) showed limited spatial overlap with the outcome map, with the highest concentrations appearing in counties where premature death rates were comparatively lower indicating a potential negative relationship. Hispanic (Appendix Figure 12) had the highest concentrations found in coastal and southern counties where premature death rates were among the lowest in the state suggesting a negative spatial association at the county level. 65 and older (Appendix Figure 13) showed a mixed spatial correspondence with premature death where northern counties with high proportions of older residents aligned with elevated premature death rates, while central inland counties with similarly high proportions of older residents did not, suggesting the relationship

between age composition and premature mortality is moderated by other county-level factors. Finally, food insecurity (Appendix Figure 14) was most concentrated in northern and central inland counties and broadly corresponded to the elevated premature death cluster in the north, though the alignment was less consistent across central inland counties.

The varied degrees of spatial correspondence between predictor and outcome maps collectively motivate the use of spatial regression models that formally account for county-level dependence rather than treating observations as independent.

4.4 OLS Regression Results

The final OLS model (Appendix Figure 4) was highly significant overall ($F = 13.53$, $p < 0.001$) and explained approximately 63.8% of county-level variation in premature death rates (Adjusted $R^2 = 0.638$). Children in poverty was the strongest and most significant predictor ($\beta = 70,271$, $p < 0.001$), with a one-unit increase in child poverty proportion associated with an increase of 70,271 YPLL per 100,000 holding all other predictors constant. Insufficient sleep was the second most significant predictor ($\beta = 57,397$, $p = 0.004$), with higher rates associated with greater premature mortality. Hispanic population share showed a significant negative association ($\beta = -7,486$, $p = 0.044$), consistent with the Hispanic health paradox. Food insecurity showed a significant but counterintuitive negative coefficient ($\beta = -80,619$, $p = 0.019$), likely reflecting suppression in the presence of the highly correlated children in poverty variable. Unemployment approached but did not reach conventional significance ($\beta = 48,025$, $p = 0.055$). Physical inactivity, Non-Hispanic Black, and 65 and older were not statistically significant at the 0.05 level.

Moran's I applied to OLS residuals (Appendix Figure 15) yielded a statistic of -0.033 ($p = 0.84$), indicating no statistically significant residual spatial autocorrelation. While this suggests the covariates largely account for the spatial structure in the outcome, spatial CAR models were nonetheless fitted to formally assess whether accounting for county-level spatial dependence alters coefficient estimates or improves model fit.

4.5 Improper CAR Results

The improper CAR model was estimated via MCMC with 200,000 burnin and 1,000,000 samples. Geweke diagnostic z-scores (Appendix Figure 16) for all regression coefficients fell within $[-2, 2]$, indicating satisfactory MCMC convergence. Effective sample sizes for all parameters (Appendix Figure 17) exceeded 1000, confirming adequate posterior exploration. The trace plots (Appendix Figure 18) for all predictors indicate good MCMC mixing and convergence with chains exhibiting stable horizontal movement across posterior means with no evidence of drift or poor mixing. As for the posterior density plots, we can see that all have a bell-shape curve with no noticeable left or right skew.

In contrast to the OLS results, our Improper CAR results (Appendix Figure 19) show that none of the regression coefficients had 95% credible intervals entirely excluding zero, indicating that no individual predictor reached statistical significance after accounting for spatially structured random effects. Coefficient estimates were substantially different relative to OLS with an example being children in poverty shrunk from $\beta = 70,271$ in OLS to a posterior median of 3,316 suggesting that the spatial random effects absorbed a considerable portion of the covariate-attributed variation, redistributing it to county-level spatial structure. Despite the loss of significance, most predictors maintained consistency with OLS estimates. Notably, food

insecurity reversed from a negative OLS coefficient to a positive posterior median, suggesting the sign reversal observed in OLS may have been a suppression artifact partially resolved by the spatial random effects. Wide credible intervals for several parameters (Ex: 65 and older) suggest high posterior uncertainty likely attributable to the limited sample of 58 counties relative to model complexity.

Moran's I applied to improper CAR residuals (Appendix Figure 20) yielded a statistic of -0.009 ($p = 0.907$), confirming that there is no residual spatial autocorrelation after fitting the model. Taken together with the non-significant Moran's I on OLS residuals ($p = 0.84$), these results suggest that spatial dependence in premature death rates across California counties is largely captured through the covariate structure, with the spatial random effects providing additional flexibility that substantially redistributes explanatory weight away from individual predictors.

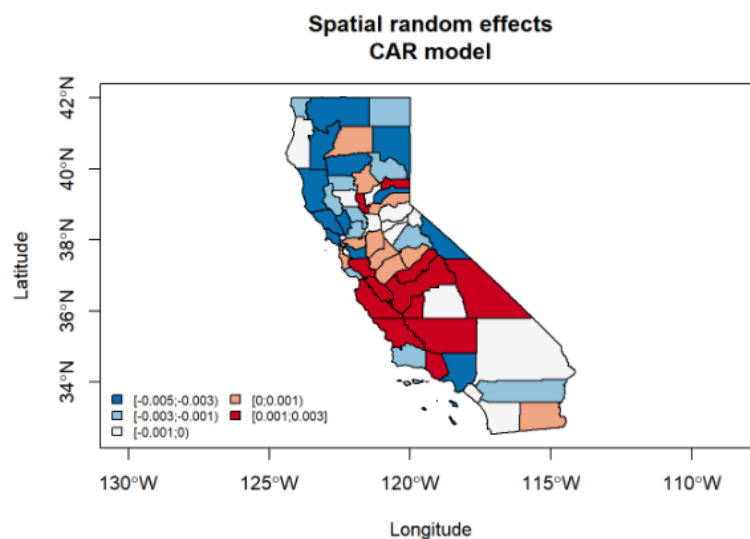


Figure 2: spatial random effects improper CAR

Figure 2 presents the posterior median spatial random effects across California's 58 counties. While the map reveals a visible geographic pattern with northern and inland counties tending toward positive random effects and coastal counties tending toward negative random effects, examination of the 95% credible intervals (Appendix figure 23 and 24) revealed that no individual county had a posterior distribution entirely above or below zero. This indicates that while suggestive spatial trends exist, there is insufficient statistical evidence to identify any specific county as having residual premature death rates higher or lower than predicted by the covariates alone. This finding is consistent with the non-significant Moran's I statistic on improper CAR residuals ($p = 0.907$), collectively suggesting that the covariate structure largely accounts for spatial dependence in premature mortality across California counties, with remaining spatial heterogeneity being modest and statistically uncertain at this sample size.

4.6 Proper CAR Results

The proper CAR model (Appendix Figure 21) estimated a spatial dependence parameter of $\lambda = -0.078$ ($SE = 0.494$), which was not statistically distinguishable from zero, indicating negligible spatial autocorrelation beyond what the covariates already captured. Consequently, proper CAR coefficient estimates closely mirrored those from OLS, with children in poverty

remaining the strongest predictor ($\beta = 66,442$, $p < 0.001$) and Hispanic population share showing a significant negative association ($\beta = -8,496$, $p = 0.018$). Notably, unemployment reached conventional significance in the proper CAR model ($\beta = 52,474$, $p = 0.031$) compared to marginal significance in OLS ($p = 0.055$), suggesting a more precise estimate once spatial structure is formally acknowledged. Food insecurity remained marginally non-significant ($\beta = -55,536$, $p = 0.077$) with a persistent negative sign. Physical inactivity, Non-Hispanic Black, and 65 and older were not significant in either model.

Moran's I on proper CAR residuals (Appendix Figure 22) yielded a statistic of -0.009 ($p = 0.909$), confirming no residual spatial autocorrelation remained after model fitting. The near-zero lambda combined with non-significant residual Moran's I suggests that the covariate structure itself largely accounts for the geographic distribution of premature death rates across California counties.

4.7 Model Comparison

	Variable	OLS Estimate (SE)	Proper CAR Estimate (SE)	Improper CAR Median (95% CrI)
Coefficients				
1	Intercept	9897.375 (4154.051)*	9880.227 (3889.71)*	-929.934 (-8151.798, 4229.931)
2	Children in Poverty	65779.27 (10920.965)***	66441.846 (10125.877)***	3315.557 (-7392.216, 13864.842)
3	Unemployment Rate	47507.07 (26270.961).	52474.233 (24377.88)*	2290.008 (-22338.104, 27129.659)
4	Physical Inactivity	-15939.404 (17835.509)	-14344.964 (16507.905)	1022.573 (-13112.347, 15197.746)
5	Non-Hispanic	-1745.683 (13849.659)	-1693.086 (12915.453)	-1026.266 (-22606.548, 20298.879)
6	Hispanic	-8072.671 (3885.773)*	-8496.447 (3595.568)*	-265.279 (-3504.229, 2963.004)
7	Aged 65+	-10183.459 (10262.387)	-10278.21 (9573.796)	1220.753 (-8578.866, 10869.42)
8	Food Insecurity	-51229.215 (34124.888)	-55535.793 (31471.637).	4478.567 (-16971.387, 27953.846)
Model Fit				
DIC	AIC / DIC	1072.12	1074.04	1214.57
	Moran's I p-value	0.84	0.9088	0.9072

Table 2: Comparison of the 3 models

Table 2 presents coefficient estimates and model fit statistics across all 3 models. In terms of model fit, all 3 produced non-significant Moran's I p-values confirming residual spatial autocorrelation was adequately addressed regardless of model.

Coefficient estimates were highly stable between OLS and proper CAR, reflecting the negligible spatial dependence captured by lambda. Children in poverty was the strongest and most consistent predictor across both models (OLS: $\beta = 65,779$, $p < 0.001$; Proper CAR: $\beta = 66,442$, $p < 0.001$), with unemployment and Hispanic population share also reaching significance in both. The improper CAR model showed no predictor reaching statistical significance, consistent with the known tendency of the intrinsic CAR prior to absorb covariate-attributed variation into spatial random effects when the sample size is small relative to model complexity.

Overall, the close agreement between OLS and proper CAR, combined with the non-significant lambda and consistent Moran's I results, suggests that the covariate structure

itself largely accounts for the spatial distribution of premature death rates across California counties, with the improper CAR providing complementary evidence of residual geographic heterogeneity through its spatial random effects.

5. Discussion

This study examined the spatial distribution of premature death rates across California's 58 counties using data from the 2025 CHR. Three models of increasing spatial complexity were compared: OLS, improper CAR, and proper CAR. Children in poverty had the largest impact across the 3 models with a highly significant and positive coefficient with premature death rate. Unemployment also had a positive coefficient and reached significance in the proper CAR model and approached significance in OLS. Physical Inactivity, Non-Hispanic Black, Hispanic, Age 65 or older, and Food Insecurity all had negative coefficients with premature death rate although only Hispanic maintained significance in either the OLS or Proper CAR estimate models suggesting true relationship with premature death is negative but difficult to isolate at this geographic resolution.

The proper CAR estimated lambda at essentially zero and achieved nearly identical AIC to OLS, indicating that the spatial clustering of premature death is largely explained by the spatial distribution of its determinants rather than unmeasured spatial processes. The improper CAR spatial random effects map revealed a suggestive pattern of residual excess mortality in northern and inland counties and a protective pattern along the coast, though no individual county's 95% credible interval for ϕ entirely excluded zero, indicating statistical uncertainty at this sample size.

Several limitations apply. With only 58 counties the models are parameter-rich relative to sample size, limiting precision of individual estimates. All inferences are ecological and cannot be extended to individuals. Additionally, the small sample size likely led to the loss of significance in the improper CAR.

6. Conclusion

This study compared OLS, proper CAR, and improper CAR models to investigate geographic determinants of premature death across California's 58 counties. Child poverty and unemployment were the most consistent socioeconomic predictors, and Hispanic population share showed a protective association. The near-zero lambda and indistinguishable AIC between OLS and proper CAR suggest that spatial clustering in the outcome is largely driven by the spatial distribution of measured covariates. The improper CAR spatial random effects map identified northern and inland counties as having residual excess mortality beyond what covariates explain, highlighting priority areas for targeted public health investment. Future work could incorporate multiple time points to improve statistical power and more sound conclusions.

7. References

“Premature Death.” County Health Rankings & Roadmaps, www.countyhealthrankings.org/health-data/population-health-and-well-being/length-of-life/life-span/premature-death?year=2025. Accessed 14 Mar. 2026.

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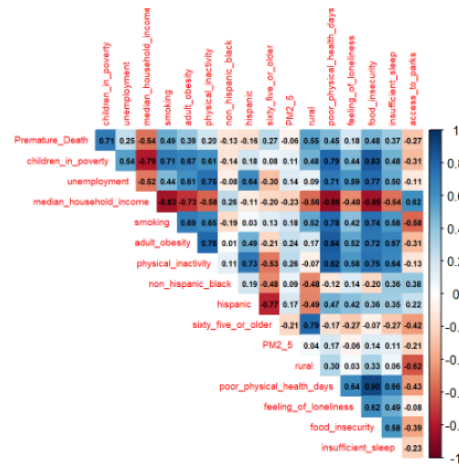
8. Appendix

Moran I test under randomisation

```
data: pre_death
weights: list
```

```
Moran I statistic standard deviate = 2.3245, p-value = 0.0201
alternative hypothesis: two.sided
sample estimates:
Moran I statistic      Expectation      Variance
0.151368941          -0.017543860          0.005280182
```

Appendix Figure 1: Moran's I test on Premature Death



Appendix Figure 2: Correlation Matrix Full Model

```
Residuals:
    Min       1Q   Median       3Q      Max
-4364.3   -821.8   -222.8    588.2   7973.0

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  2.410e+04  1.886e+04  1.278  0.208519
children_in_poverty  5.662e+04  1.392e+04  4.067  0.000211 ***
unemployment      3.962e+04  2.975e+04  1.331  0.190393
median_household_income -3.945e+02  4.476e+02  -0.882  0.383167
smoking          3.812e+03  4.081e+04  0.096  0.924093
adult_obesity     1.418e+04  1.499e+04  0.946  0.349747
physical_inactivity -4.322e+04  3.164e+04  -1.366  0.179336
non_hispanic_black -1.403e+04  1.881e+04  -0.746  0.460048
hispanic          -5.676e+03  5.984e+03  -0.948  0.348454
sixty_five_or_older -3.942e+04  1.588e+04  -2.483  0.017229 *
PM2_5           -1.364e+02  8.002e+01  -1.705  0.095852 .
rural            4.909e+03  3.041e+03  1.614  0.114147
poor_physical_health_days -1.811e+03  2.504e+03  -0.723  0.473586
feeling_of_loneliness -2.079e+04  2.932e+04  -0.709  0.482257
food_insecurity    -5.436e+04  4.895e+04  -1.111  0.273226
insufficient_sleep  3.959e+04  2.320e+04  1.706  0.095502 .
access_to_parks    6.170e+02  1.972e+03  0.313  0.755950
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2216 on 41 degrees of freedom
Multiple R-squared:  0.7202, Adjusted R-squared:  0.6111
F-statistic: 6.597 on 16 and 41 DF,  p-value: 5.376e-07
```

Appendix Figure 3: Full Model Summary Statistics

```

Residuals:
    Min       1Q   Median       3Q      Max
-3422.7 -1082.6  -327.8   867.1  8969.1

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)    -2451      5642   -0.434  0.66591
children_in_poverty  70271    10250   6.855 1.11e-08 ***
unemployment    48025     24394   1.969  0.05465 .
physical_inactivity -31041    17309  -1.793  0.07910 .
non_hispanic_black -27059    15382  -1.759  0.08480 .
hispanic        -7486     3613   -2.072  0.04357 *
sixty_five_or_older -14885    9657  -1.541  0.12967
food_insecurity  -80619    33167  -2.431  0.01877 *
insufficient_sleep  57397    19139   2.999  0.00425 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2140 on 49 degrees of freedom
Multiple R-squared:  0.6594, Adjusted R-squared:  0.6376
F-statistic: 13.53 on 8 and 49 DF,  p-value: 4.171e-10

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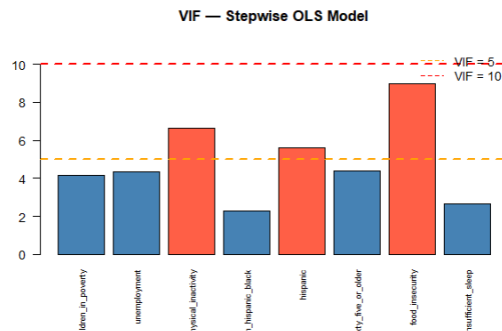
Appendix Figure 4: Reduced Model Summary Statistics

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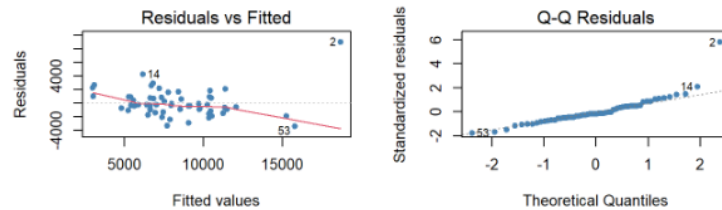
## Analysis of Variance Table
##
## Model 1: Premature_Death ~ children_in_poverty + unemployment + physical_inactivity +
##   non_hispanic_black + hispanic + sixty_five_or_older + food_insecurity +
##   insufficient_sleep
## Model 2: Premature_Death ~ children_in_poverty + unemployment + median_household_income +
##   smoking + adult_obesity + physical_inactivity + non_hispanic_black +
##   hispanic + sixty_five_or_older + PM2_5 + rural + poor_physical_health_days +
##   feeling_of_loneliness + food_insecurity + insufficient_sleep +
##   access_to_parks
## Res.Df    RSS Df Sum of Sq    F Pr(>F)
## 1      49 224300497
## 2     41 201402880  8  22897617 0.5827 0.7861

```

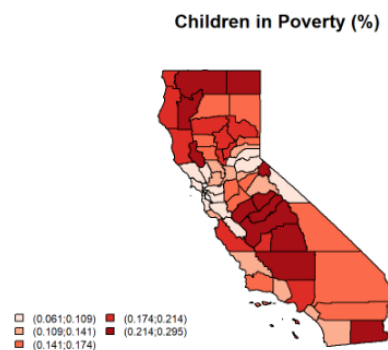
Appendix Figure 5: Anova test of full OLS vs reduced OLS model



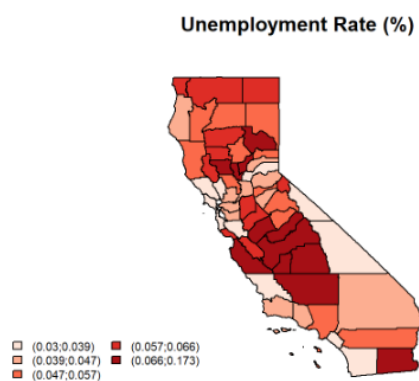
Appendix Figure 6: VIF Analysis on Reduced Model



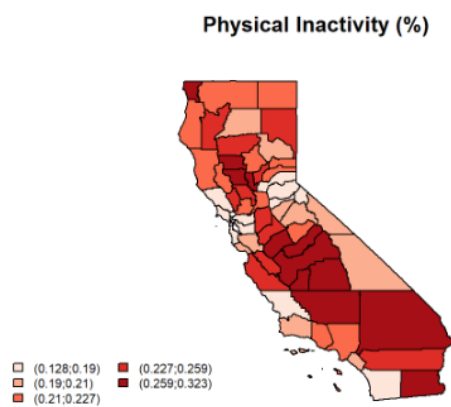
Appendix Figure 7: Model Diagnostics on Reduced Model



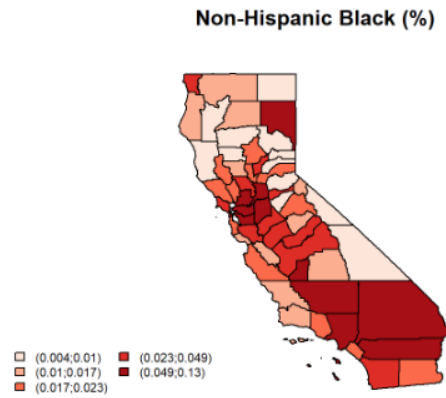
Appendix Figure 8: Choropleth Map of Covariate: Children in Poverty



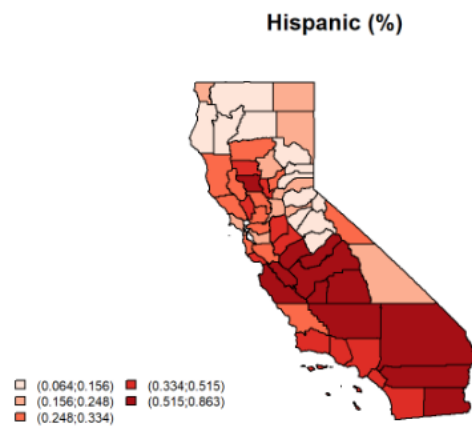
Appendix Figure 9: Choropleth Map of Covariate: Unemployment Rate



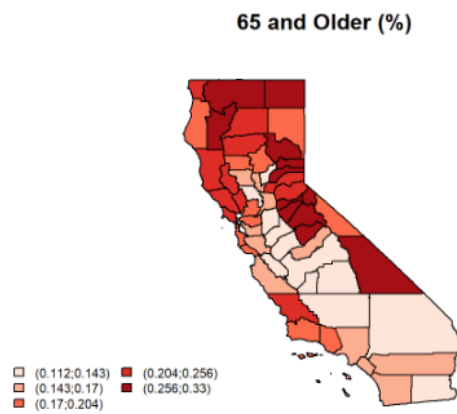
Appendix Figure 10: Choropleth Map of Covariate: Physical Inactivity



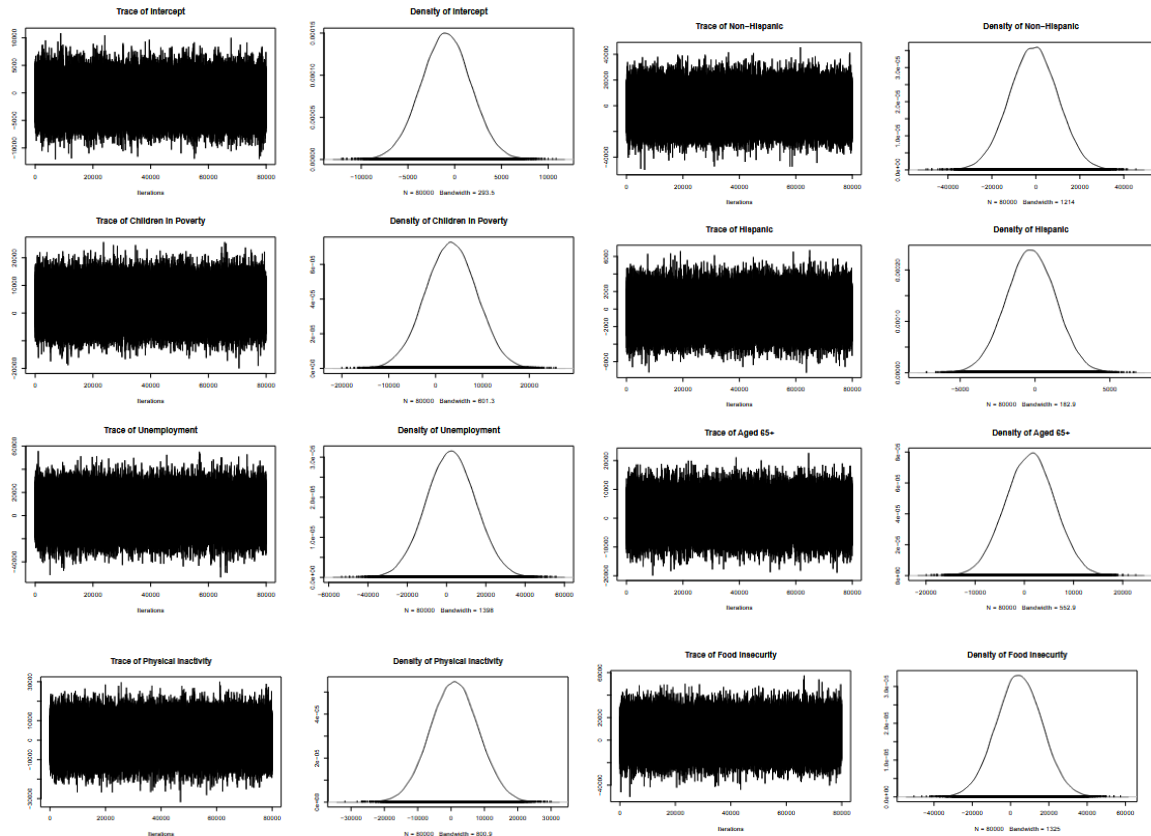
Appendix Figure 11: Choropleth Map of Covariate: Non-Hispanic Black



Appendix Figure 12: Choropleth Map of Covariate: Hispanic



Appendix Figure 13: Choropleth Map of Covariate: 65 and Older



Appendix Figure 18: Trace and Density Plots of Covariates in Improper CAR

Intercept Children in Poverty		Unemployment Physical Inactivity		Non-Hispanic		Hispanic		Aged 65+		Food Insecurity	
-929.9342 3315.5566		2290.0082 1022.5730		-1026.2660		-265.2786		1220.7532		4478.5667	
Intercept Children in Poverty Unemployment Physical Inactivity		-13112.35 -22606.55		Hispanic Aged 65+ Food Insecurity		-3504.229 -8578.866		-18971.39			
2.5% -6151.798		-7392.216 -22338.10		2963.004 10869.420		27953.85					
97.5% 4229.931		13864.842 27129.66		15197.75 20298.88							

Appendix Figure 19: Table of Median and 95% Confidence Intervals for Improper CAR Covariates

```

Moran I test under randomisation

data: residuals(car)
weights: list

Moran I statistic standard deviate = 0.11659, p-value = 0.9072
alternative hypothesis: two.sided
sample estimates:
Moran I statistic      Expectation      Variance
-0.009375097          -0.017543860          0.004909299

```

Appendix Figure 20: Moran's I for Improper CAR

```

Lambda (rho) = -0.0784 SE = 0.4941
      Estimate Std. Error z value Pr(>|z|)
(Intercept)    9860.227   3869.710   2.5481  0.0108
children_in_poverty 66441.846 10125.677   6.5617  0.0000
unemployment    52474.233 24377.880   2.1525  0.0314
physical_inactivity -14344.964 16507.906  -0.8690  0.3849
non_hispanic_black -1693.086 12915.452  -0.1311  0.8957
hispanic        -8496.447  3595.568  -2.3630  0.0181
sixty_five_or_older -10278.210  9573.796  -1.0736  0.2830
food_insecurity  -55535.793 31471.637  -1.7646  0.0776

```

Appendix Figure 21: Summary Statistics for proper CAR model

```

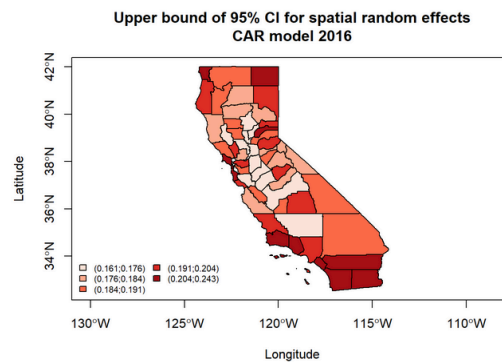
Moran I test under randomisation

data: residuals(proper_car)
weights: list

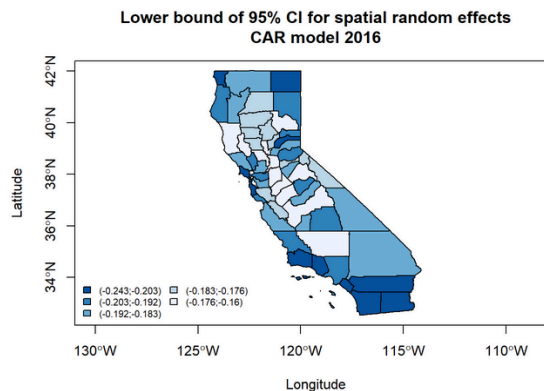
Moran I statistic standard deviate = 0.1148, p-value = 0.9086
alternative hypothesis: two.sided
sample estimates:
Moran I statistic      Expectation      Variance
-0.008506728          -0.017543860        0.006197121

```

Appendix Figure 22: Moran's I for Proper CAR



Appendix Figure 23: Upper bound 95% CI Improper CAR



Appendix Figure 24: Lower bound 95% CI Improper CAR

9. Code

```
library(readr)
library(sf)
library(maps)
library(spdep)
library(RColorBrewer)
library(tidyverse)
library(MASS)
library(car)
library(tigris)
library(CARBayes)
library(coda)
library(ggplot2)
library(classInt)
library(car)
library(spatialreg)
library(knitr)
library(kableExtra)
select = dplyr::select
filter = dplyr::filter
rename = dplyr::rename
mutate = dplyr::mutate
analytic_data2025_v3 = read_csv("C:/Users/Cameron Fong/Downloads/analytic_data2025_v3.csv", skip
= 1)
```

```{r}
#obtaining california data
california_data = analytic_data2025_v3 %>% filter(state %in% "CA") %>% filter(county != "California")
california_county_names = california_data$county
california_data = california_data %>% dplyr::rename(
  Premature_Death = v001_rawvalue,
  children_in_poverty = v024_rawvalue,
  unemployment = v023_rawvalue,
  median_household_income = v063_rawvalue,
  smoking = v009_rawvalue,
  adult_obesity = v011_rawvalue,
  physical_inactivity = v070_rawvalue,
  non_hispanic_black = v054_rawvalue,
  hispanic = v056_rawvalue,
  sixty_five_or_older = v053_rawvalue,
  PM2_5 = v125_rawvalue,
```

```

rural = v058_rawvalue,
poor_physical_health_days = v036_rawvalue,
feeling_of_loneliness = v183_rawvalue,
food_insecurity = v139_rawvalue,
insufficient_sleep = v143_rawvalue,
access_to_parks = v179_rawvalue
)

```

```

#keeping variables of interest

```

```

keep_vars = c("fipscode",
              "Premature_Death",
              "children_in_poverty",
              "unemployment",
              "median_household_income",
              "smoking",
              "adult_obesity",
              "physical_inactivity",
              "non_hispanic_black",
              "hispanic",
              "sixty_five_or_older",
              "PM2_5", # PM2.5
              "rural", # rural
              "poor_physical_health_days",
              "feeling_of_loneliness",
              "food_insecurity",
              "insufficient_sleep",
              "access_to_parks")

```

```

california_data = california_data[, keep_vars]

```

```

ca_counties = counties(state = "CA", cb = TRUE, year = 2023) %>%
  st_transform(crs = 4326)

```

```

california_data = california_data %>%
  mutate(fips = str_pad(as.character(fipscode), 5, pad = "0"))

```

```

# Join geometry to CHR data

```

```

ca_sf = left_join(ca_counties, california_data, by = c("GEOID" = "fips"))

```

```

california_data = california_data %>%
  select(-fipscode, -fips)

```

```

#choropleth map for outcome variable premature death

```

```

plotvar = ca_sf$Premature_Death

```

```

nclr = 5
plotclr = brewer.pal(nclr, "YlOrRd")
class = classIntervals(plotvar, nclr, style = "fixed",
                        fixedBreaks = quantile(plotvar,
                        c(0, 0.2, 0.4, 0.6, 0.8, 1.0), na.rm = TRUE))
colcode = findColours(class, plotclr)
plot(st_geometry(ca_sf), border = "black", axes = FALSE)
title(xlab = "Longitude", ylab = "Latitude",
      main = "Premature Death Rate by California County (2025)")
plot(st_geometry(ca_sf), col = colcode, add = TRUE)
leg.txt = paste0("(", round(class$brks[-length(class$brks)], 3),
                  ":", round(class$brks[-1], 3), ")")
legend("bottomleft", legend = leg.txt, fill = plotclr, cex = 0.7, ncol = 2, bty = "n")

```

```

sf_use_s2(FALSE)
nb = poly2nb(ca_sf, queen = TRUE)
weights = nb2WB(nb)
adj.weights = weights$adj
weights.nation = weights$weights
num = weights$num
list = nb2listw(nb, zero.policy = TRUE)

```

```

#Moran's I for 2016
pre_death = ca_sf$Premature_Death
moran = moran.test(pre_death, listw = list,
                   zero.policy = TRUE, alternative = "two.sided")
moran

```

```

#full ols model
ols_full = lm(Premature_Death ~ .,
              data = california_data)
summary(ols_full)

```

```

#correlation matrix
cor_matrix = cor(california_data)
print(round(cor_matrix, 2))

```

```

corrplot::corrplot(cor_matrix, method = "color", type = "upper",
                   tl.cex = 0.6, addCoef.col = "black",
                   number.cex = 0.5, diag = FALSE,
                   title = "Correlation Matrix — Full Variable Set",
                   mar = c(0, 0, 1, 0))

```

```

#stepwise AIC

```

```

ols_step = stepAIC(ols_full, direction = "both", trace = TRUE)
summary(ols_step)

#anova test full vs stepwise regression result
anova(ols_step, ols_full)

#residual diagnostics - VIF
par(mfrow = c(2, 2))
plot(ols_step, pch = 19, col = "steelblue", cex = 0.8)
par(mfrow = c(1, 1))

vif_vals = vif(ols_step)
print(round(vif_vals, 3))

barplot(vif_vals,
        main = "VIF — Stepwise OLS Model",
        col = ifelse(vif_vals > 5, "tomato", "steelblue"),
        las = 2, cex.names = 0.65,
        ylim = c(0, max(vif_vals) + 2))
abline(h = 5, col = "orange", lty = 2, lwd = 2)
abline(h = 10, col = "red", lty = 2, lwd = 2)
legend("topright", legend = c("VIF = 5", "VIF = 10"),
      col = c("orange", "red"), lty = 2, bty = "n")

#choropleth map of predictor variables
plot_choropleth = function(sf_obj, variable, title) {
  plotvar = sf_obj[[variable]]
  nclr = 5
  plotclr = brewer.pal(nclr, "Reds")
  class = classIntervals(plotvar, nclr, style = "fixed",
    fixedBreaks = quantile(plotvar,
    c(0, 0.2, 0.4, 0.6, 0.8, 1.0), na.rm = TRUE))
  colcode = findColours(class, plotclr)
  leg.txt = paste0("(", round(class$brks[-length(class$brks)], 3),
    ":", round(class$brks[-1], 3), ")")
  plot(st_geometry(sf_obj), border = "black", axes = FALSE)
  title(xlab = "Longitude", ylab = "Latitude", main = title)
  plot(st_geometry(sf_obj), col = colcode, add = TRUE)
  legend("bottomleft", legend = leg.txt, fill = plotclr,
    cex = 0.7, ncol = 2, bty = "n")
}

plot_choropleth(ca_sf, "children_in_poverty", "Children in Poverty (%)")
plot_choropleth(ca_sf, "unemployment", "Unemployment Rate (%)")

```

```

plot_choropleth(ca_sf, "physical_inactivity", "Physical Inactivity (%)")
plot_choropleth(ca_sf, "non_hispanic_black", "Non-Hispanic Black (%)")
plot_choropleth(ca_sf, "hispanic", "Hispanic (%)")
plot_choropleth(ca_sf, "sixty_five_or_older", "65 and Older (%)")
plot_choropleth(ca_sf, "food_insecurity", "Food Insecurity (%)")

#final model chosen
final_model = lm(Premature_Death ~ children_in_poverty + unemployment + physical_inactivity +
non_hispanic_black + hispanic + sixty_five_or_older + food_insecurity, data = california_data)

# Variables in final model
desc_vars = c("Premature_Death", "children_in_poverty", "unemployment",
              "physical_inactivity", "non_hispanic_black", "hispanic",
              "sixty_five_or_older", "food_insecurity")

#Table one
var_labels = c(
  Premature_Death = "Premature Death Rate (YPLL/100k)",
  children_in_poverty = "Children in Poverty (%)",
  unemployment = "Unemployment Rate (%)",
  physical_inactivity = "Physical Inactivity (%)",
  non_hispanic_black = "Non-Hispanic Black (%)",
  hispanic = "Hispanic (%)",
  sixty_five_or_older = "65 and Older (%)",
  food_insecurity = "Food Insecurity (%)"
)

desc_table = california_data[, desc_vars] %>%
  pivot_longer(everything(), names_to = "Variable") %>%
  group_by(Variable) %>%
  summarise(
    N = sum(!is.na(value)),
    Mean = round(mean(value, na.rm = TRUE), 3),
    SD = round(sd(value, na.rm = TRUE), 3),
    Min = round(min(value, na.rm = TRUE), 3),
    Q25 = round(quantile(value, 0.25, na.rm = TRUE), 3),
    Median = round(median(value, na.rm = TRUE), 3),
    Q75 = round(quantile(value, 0.75, na.rm = TRUE), 3),
    Max = round(max(value, na.rm = TRUE), 3),
    .groups = "drop"
  )
desc_table$Variable = var_labels[desc_table$Variable]

print(kable(desc_table,

```

```

caption = "Table 1: Descriptive Statistics — California Counties (2025 CHR, n=58)",
align = c("l", rep("r", 8)))

#Moran's I OLS model
moran.test(residuals(final_model), list,
           zero.policy = TRUE, alternative = "two.sided")

#Improper CAR

W_mat = nb2mat(nb, style = "B", zero.policy = TRUE)

car = S.CARleroux(formula = Premature_Death ~ children_in_poverty + unemployment +
physical_inactivity + non_hispanic_black + hispanic + sixty_five_or_older + food_insecurity, data =
california_data, family = "gaussian", W = W_mat, burnin = 200000, n.sample = 1000000, thin = 10,
verbose = TRUE)

#improper CAR model diagnostics
colnames(car$samples$beta) = c("Intercept", "Children in Poverty",
                             "Unemployment", "Physical Inactivity",
                             "Non-Hispanic", "Hispanic",
                             "Aged 65+", "Food Insecurity")
geweke.diag(car$samples$beta)
colnames(car$samples$phi) = california_county_names
geweke.diag(car$samples$phi)

car$summary.results

beta_coda = as.mcmc(car$samples$beta)
colnames(beta_coda) = c("Intercept", "Children in Poverty",
                       "Unemployment", "Physical Inactivity",
                       "Non-Hispanic", "Hispanic",
                       "Aged 65+", "Food Insecurity")
pdf("mcmc_diagnostics.pdf", width = 10, height = 14)
plot(beta_coda)

#plotting spatial effects

phi_samples = car$samples$phi

phi_median = as.numeric(apply(phi_samples, 2, median))
ca_sf$phi_median = phi_median

plotvar = ca_sf$phi_median
nclr = 5

```

```

plotclr = brewer.pal(nclr, "RdBu")[5:1]
class = classIntervals(plotvar, nclr, style = "quantile")

colcode = findColours(class, plotclr)
plot(st_geometry(ca_sf), border = "black", axes = TRUE)
title(xlab = "Longitude", ylab = "Latitude",
      main = "Spatial random effects\nCAR model")
plot(st_geometry(ca_sf), col = colcode, add = TRUE)
leg.txt = c("[-0.005;-0.003)", "[-0.003;-0.001)", "[-0.001;0)",
            "[0;0.001)", "[0.001;0.003]")
legend("bottomleft", legend = leg.txt, fill = plotclr, cex = 0.7, ncol = 2, bty = "n")

#Lower bound 95% CI
low95 = as.numeric(apply(phi_samples, 2, quantile, 0.025))
ca_sf$phi_low95 = low95

plotvar = ca_sf$phi_low95
nclr = 5
plotclr = brewer.pal(nclr, "Blues")[5:1]
class = classIntervals(plotvar, nclr, style = "fixed",
                      fixedBreaks = quantile(plotvar,
                      c(0, 0.2, 0.4, 0.6, 0.8, 1.0), na.rm = TRUE))
colcode = findColours(class, plotclr)
plot(st_geometry(ca_sf), border = "black", axes = TRUE)
title(xlab = "Longitude", ylab = "Latitude",
      main = "Lower bound of 95% CI for spatial random effects\nCAR model 2016")
plot(st_geometry(ca_sf), col = colcode, add = TRUE)
leg.txt = paste0("(", round(class$brks[-length(class$brks)], 3),
                ":", round(class$brks[-1], 3), ")")
legend("bottomleft", legend = leg.txt, fill = plotclr, cex = 0.7, ncol = 2, bty = "n")

#Upper bound 95% CI
up95 = as.numeric(apply(phi_samples, 2, quantile, 0.975))
ca_sf$phi_up95 = up95

plotvar = ca_sf$phi_up95
nclr = 5
plotclr = brewer.pal(nclr, "Reds")
class = classIntervals(plotvar, nclr, style = "fixed",
                      fixedBreaks = quantile(plotvar,
                      c(0, 0.2, 0.4, 0.6, 0.8, 1.0), na.rm = TRUE))
colcode = findColours(class, plotclr)
plot(st_geometry(ca_sf), border = "black", axes = TRUE)
title(xlab = "Longitude", ylab = "Latitude",

```

```

main = "Upper bound of 95% CI for spatial random effects\nCAR model 2016")
plot(st_geometry(ca_sf), col = colcode, add = TRUE)
leg.txt = paste0("(", round(class$brks[-length(class$brks)], 3),
                ":", round(class$brks[-1], 3), ")")
legend("bottomleft", legend = leg.txt, fill = plotclr, cex = 0.7, ncol = 2, bty = "n")

#95% CI bounds for all states
ci_bounds = apply(phi_samples, 2, quantile, probs = c(0.025, 0.975))
colnames(ci_bounds) = ca_sf$NAME
print(round(ci_bounds, 4))

#Identify states with entirely positive or negative phi
pos_states = ca_sf$NAME[low95 > 0]
neg_states = ca_sf$NAME[up95 < 0]
cat("\nStates with phi 95% CI entirely POSITIVE:\n")
if (length(pos_states) == 0) cat(" None\n") else print(pos_states)
cat("\nStates with phi 95% CI entirely NEGATIVE:\n")
if (length(neg_states) == 0) cat(" None\n") else print(neg_states)

#output of beta values and 95% CI
beta_samples = car$samples$beta
apply(beta_samples, 2, median)
apply(beta_samples, 2, quantile, probs = c(0.025, 0.975))

#moran's I test for improper CAR
moran.test(residuals(car),list,zero.policy = TRUE,alternative = "two.sided")

#proper CAR
proper_car = spautolm(Premature_Death ~ children_in_poverty + unemployment + physical_inactivity +
non_hispanic_black + hispanic + sixty_five_or_older + food_insecurity, data =
california_data,listw=list,family="CAR")

coefs = summary(proper_car)$Coef
print("Proper CAR model")
cat(sprintf("Lambda (rho) = %.4f SE = %.4f\n",
            proper_car$lambda, proper_car$lambda.se))
print(round(coefs, 4))

#moran's I for proper CAR
moran.test(residuals(proper_car),list,zero.policy = TRUE,alternative = "two.sided")

#3 model comparison table

#obtaining OLS values

```

```

ols_coef = summary(final_model)$coefficients
ols_df = data.frame(
  Estimate = ols_coef[, "Estimate"],
  SE       = ols_coef[, "Std. Error"],
  p        = ols_coef[, "Pr(>|t|)"]
)
ols_df$stars = ifelse(ols_df$p < 0.001, "****",
  ifelse(ols_df$p < 0.01, "***",
    ifelse(ols_df$p < 0.05, "**",
      ifelse(ols_df$p < 0.1, ".", ""))))
ols_df$cell = paste0(round(ols_df$Estimate, 3),
  " (", round(ols_df$SE, 3), ")",
  ols_df$stars)

#obtaining proper CAR values
pcar_coef = summary(proper_car)$Coef
pcar_df = data.frame(
  Estimate = pcar_coef[, "Estimate"],
  SE       = pcar_coef[, "Std. Error"],
  p        = pcar_coef[, "Pr(>|z|)"]
)
pcar_df$stars = ifelse(pcar_df$p < 0.001, "****",
  ifelse(pcar_df$p < 0.01, "***",
    ifelse(pcar_df$p < 0.05, "**",
      ifelse(pcar_df$p < 0.1, ".", ""))))
pcar_df$cell = paste0(round(pcar_df$Estimate, 3),
  " (", round(pcar_df$SE, 3), ")",
  pcar_df$stars)

#obtaining improper CAR values
beta_med = apply(car$samples$beta, 2, median)
beta_lo  = apply(car$samples$beta, 2, quantile, 0.025)
beta_hi  = apply(car$samples$beta, 2, quantile, 0.975)

# Mark significance: 95% CrI excludes 0
icar_sig = ifelse(beta_lo > 0 | beta_hi < 0, "*", "")

icar_cell = paste0(round(beta_med, 3),
  " (", round(beta_lo, 3),
  ", ", round(beta_hi, 3), ")",
  icar_sig)

#row labels
var_names = c("Intercept", "Children in Poverty", "Unemployment Rate",

```

```

      "Physical Inactivity", "Non-Hispanic", "Hispanic",
      "Aged 65+", "Food Insecurity")

#putting table together
model_table = data.frame(
  Variable = var_names,
  OLS      = ols_df$cell,
  Proper_CAR = pcar_df$cell,
  Improper_CAR = icar_cell,
  stringsAsFactors = FALSE
)

#Adding moran's I tests
ols_moran = moran.test(residuals(final_model), list,
  zero.policy = TRUE, alternative = "two.sided")
pcar_moran = moran.test(residuals(proper_car), list,
  zero.policy = TRUE, alternative = "two.sided")
icar_moran = moran.test(residuals(car), list,
  zero.policy = TRUE, alternative = "two.sided")

fit_rows = data.frame(
  Variable = c("AIC / DIC",
    "Moran's I p-value"),
  OLS      = c(round(AIC(final_model), 2),
    round(ols_moran$p.value, 4)),
  Proper_CAR = c(round(AIC(proper_car), 2),
    round(pcar_moran$p.value, 4)),
  Improper_CAR = c(round(car$model$fit["DIC"], 2),
    round(icar_moran$p.value, 4)),
  stringsAsFactors = FALSE
)

model_table = rbind(model_table, fit_rows)

#printing of table
kable(model_table,
  col.names = c("Variable",
    "OLS Estimate (SE)",
    "Proper CAR Estimate (SE)",
    "Improper CAR Median (95% CrI)"),
  caption = "Table 2: Model Comparison — OLS, Proper CAR, and Improper CAR",
  align = c("l", "c", "c", "c"),
  booktabs = TRUE) %>%
kable_styling(full_width = FALSE,

```

```
        latex_options = c("hold_position", "scale_down"),
        font_size = 11) %>%
row_spec(0, bold = TRUE, background = "#D9E1F2") %>%
row_spec(nrow(model_table) - 3, # line before fit stats
        extra_css = "border-top: 2px solid black;") %>%
pack_rows("Coefficients", 1, 8) %>%
pack_rows("Model Fit", 9, 9)
```